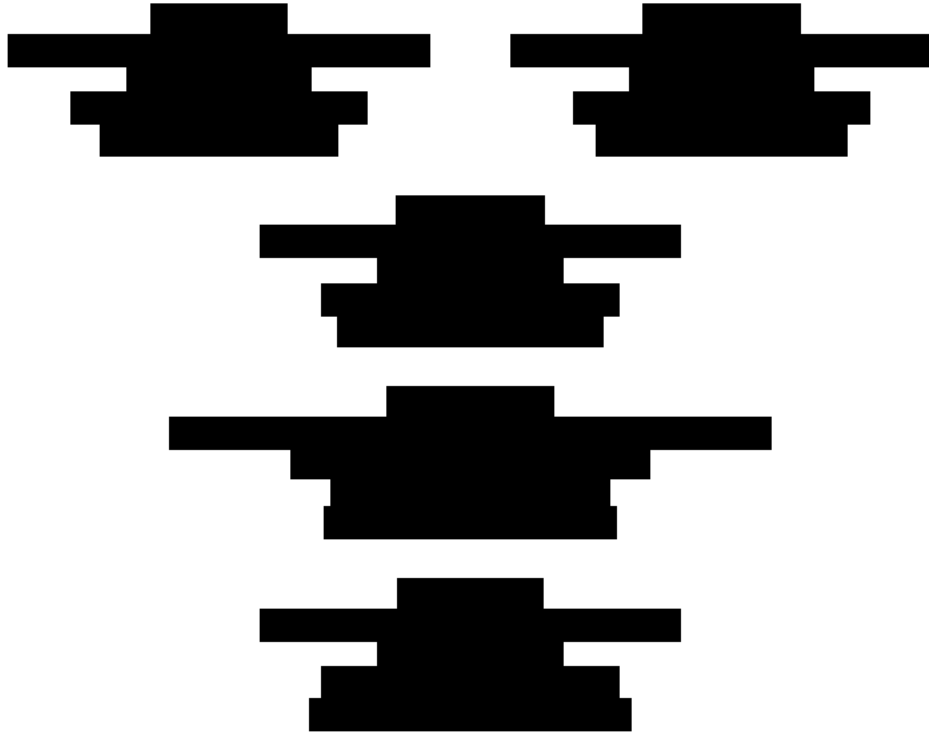


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# SparseUrban: Conditional DDIM for Turbulent Flow Field Recovery

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GPT-5.2, Gemini 3 Pro and Claude Opus 4.5



## Abstract

Accurate prediction of urban airflow is essential for addressing critical challenges in smart city applications, including pollutant dispersion monitoring and pedestrian wind comfort assessment. While high-fidelity Computational Fluid Dynamics (CFD) simulations, particularly Large Eddy Simulation (LES), can resolve turbulent flow structures with high accuracy, their prohibitive computational cost limits real-time deployment. Existing data-driven surrogate models based on deterministic regression suffer from spectral bias, producing over-smoothed predictions that attenuate high-frequency turbulent fluctuations. In this work, we propose a conditional Denoising Diffusion Probabilistic Model (DDPM) framework for reconstructing dense urban velocity fields from extremely sparse sensor measurements (5% observation ratio). Our approach incorporates physics-aware design choices including signed distance function (SDF) encoding for building geometry, interpolation-based input preprocessing, and a residual diffusion formulation that focuses learning on high-frequency corrections. Systematic evaluation on the UrbanTALES dataset across four diverse urban configurations demonstrates

that the proposed model achieves  $R^2$  scores of 0.546–0.671 for flow speed reconstruction, significantly outperforming deterministic U-Net baselines (0.355–0.540). The results confirm that diffusion-based generative modeling effectively recovers multi-scale turbulent structures, including complex building wake regions and canyon channeling effects, establishing a promising direction for real-time urban flow prediction under sparse observational constraints.

**Keywords:** Denoising Diffusion Probabilistic Model (DDPM), Urban Flow Reconstruction, Sparse Sensor Reconstruction, Signed Distance Function (SDF), UrbanTALES Dataset, Computational Fluid Dynamics (CFD), Deep Learning

## 1 Introduction

The unprecedented pace of global urbanization has rendered high-fidelity urban airflow modeling an indispensable tool for addressing critical societal challenges, including pollutant dispersion monitoring, urban heat island mitigation, and pedestrian wind comfort assessment. Accurate predictions of velocity fields and turbulent statistics within complex urban canopies are traditionally achieved through Computational Fluid Dynamics (CFD), particularly Large Eddy Simulation (LES), which resolves the energy-containing turbulent eddies responsible for momentum and scalar transport García-Sánchez et al. [2018]. However, the extreme computational cost of such high-fidelity simulations—often requiring millions of CPU hours for a single urban district—prohibits their deployment in real-time decision-making, iterative urban design optimization, and large-scale climate adaptation planning that demand rapid feedback cycles Jiménez [2003], Mortezaazadeh et al. [2022].

To bridge this gap between physical fidelity and computational tractability, the research community has increasingly turned to data-driven surrogate models that leverage the representational power of deep neural networks. Pioneering work demonstrated that convolutional neural networks (CNNs) could predict unsteady flow fields around bluff bodies with remarkable accuracy, establishing a paradigm for learning complex spatiotemporal fluid dynamics from simulation data Lee and You [2019]. Subsequent advances have expanded this foundation considerably: super-resolution frameworks capable of reconstructing fine-scale turbulent structures from coarse observations have been developed Fukami et al. [2019, 2020], while CNNs have been employed to identify drag-inducing roughness elements on complex surfaces Shin et al. [2024]. The broader integration of machine learning into computational fluid mechanics has been critically assessed Taira et al. [2025] and comprehensively reviewed Vinuesa and Brunton [2022], articulating both the transformative potential and the methodological challenges inherent in this rapidly evolving field.

Within the specific domain of urban flow prediction, several neural network architectures have been proposed to address the unique challenges posed by irregular building geometries and multi-scale turbulent interactions. PIGNN-CFD, a physics-informed graph neural network designed to handle the unstructured mesh representations common in urban CFD, incorporates Reynolds-Averaged Navier-Stokes (RANS) constraints into the loss function to enforce physical consistency Shao et al. [2023]. Similarly, graph-assisted autoencoders for sparse sensor reconstruction have demonstrated that limited observational data could be leveraged to infer dense flow fields across urban domains Gao et al. [2024]. While these graph-based approaches offer promising scalability to larger urban extents, they remain fundamentally constrained by their deterministic regression objectives, which minimize mean squared error and consequently produce smoothed predictions that attenuate high-frequency turbulent fluctuations.

This spectral bias of deterministic models poses a critical limitation for urban flow applications. Previous studies have emphasized that LES captures turbulent fluctuations with substantially greater fidelity than RANS-based methods, underscoring the importance of preserving multi-scale flow structures in any surrogate model García-Sánchez et al. [2018]. The physical significance of these fluctuations—governing phenomena from pedestrian-level gust events to pollutant concentration peaks—cannot be adequately represented by averaged or smoothed predictions. Generative adversarial networks (GANs) have been explored as a potential remedy, achieving improved structural similarity in early-stage urban design applications Kastner and Dogan [2023]. However, GAN-based approaches suffer from well-documented training instabilities and mode collapse, particularly when applied to the complex wake regions characteristic of dense urban morphologies.

The emergence of denoising diffusion probabilistic models (DDPMs) represents a paradigm shift in generative modeling that addresses many limitations of prior approaches Ho et al. [2020]. By learning to reverse a gradual noising process, diffusion models achieve state-of-the-art sample quality across diverse domains while maintaining training stability far superior to adversarial methods Dhariwal and Nichol [2021]. The impact of diffusion-based architectures has been transformative across computer vision and generative AI more broadly, as evidenced by their integration into frontier multimodal systems such as Gemini, which leverages diffusion principles for high-fidelity image generation Comanici et al. [2025]. Crucially for physical applications, the iterative refinement process inherent to diffusion sampling naturally recovers fine-grained details that deterministic regression tends to suppress, making these models particularly well-suited for turbulent flow reconstruction where multi-scale fidelity is paramount.

The availability of high-quality benchmark datasets has further accelerated progress in urban flow modeling. The UrbanTALES project provides rigorously validated LES data for over 500 distinct urban configurations worldwide, establishing the largest open-access repository for urban canopy layer turbulence research Nazarian et al. [2025]. This unprecedented data resource enables systematic evaluation of surrogate models across diverse morphological characteristics, from dense high-rise districts to sparse suburban layouts, facilitating the development of generalizable approaches rather than site-specific solutions.

In this work, we introduce a conditional diffusion framework specifically designed for reconstructing dense urban flow fields from sparse sensor measurements. By formulating the reconstruction task as conditional generation rather than deterministic regression, our approach learns the full probability distribution over plausible flow states given limited observations, naturally capturing the inherent uncertainty and multi-scale variability of urban turbulence. The framework incorporates several design choices motivated by the physics of urban flows: signed distance function (SDF) encoding of building geometry to provide continuous boundary condition information, interpolation-based input preprocessing to align sparse measurements with convolutional inductive biases, and residual diffusion formulation to focus learning capacity on high-frequency corrections where deterministic methods fail.

The primary contributions of this paper are threefold:

- We develop a conditional DDPM framework for high-resolution urban flow reconstruction ( $u, v$  velocity components) from extremely sparse sensor measurements (5% observation ratio), demonstrating that generative modeling can overcome the spectral bias inherent in deterministic regression approaches.
- We provide systematic comparison against baseline methods (linear interpolation, U-Net) across multiple urban configurations from the UrbanTALES dataset, quantifying improvements in both global error metrics and the recovery of turbulent flow structures near complex building geometries.
- We demonstrate that the proposed residual diffusion formulation, combined with physics-aware input encoding, enables effective reconstruction of wake regions and channeling effects that are critical for practical urban flow applications yet poorly captured by conventional surrogate models.

The remainder of this paper is organized as follows. Section 2 describes the dataset, preprocessing pipeline, and sparse sensor emulation methodology. Section 3 presents the diffusion-based reconstruction framework and key design choices. Section 4 reports experimental results across four urban test cases, and Section 5 concludes with discussion of limitations and future directions.

## 2 Dataset

### 2.1 Data description

The ground-truth data for this study is sourced from the *UrbanTALES* project Nazarian et al. [2025] developed by the UNSW CRC Lab, which provides high-resolution Large Eddy Simulation (LES) data derived from actual urban morphologies. To ensure the model captures the complex physical interactions present in diverse real-world environments, we employed the entire ensemble of 314 distinct realistic urban configurations for model training.

By utilizing all 314 cases, the model is exposed to a wide spectrum of building densities and vertical heterogeneities, enabling it to learn universal physical laws of urban airflow rather than overfitting to specific site patterns. For all cases within this dataset, a consistent inflow wind direction of  $0^\circ$  was maintained, providing a standardized framework for the model to learn to reconstruct flow fields based on local geometric features common to any urban environment.

## 2.2 Geometric Representation via Signed Distance Function (SDF)

To encode the intricate boundary conditions of urban structures, a Signed Distance Function (SDF) is employed. The SDF, denoted as  $\Phi(\mathbf{x})$ , is calculated as the signed Euclidean distance from a point  $\mathbf{x}$  to the nearest building boundary:

$$\Phi(\mathbf{x}) = d_{out}(\mathbf{x}) - d_{in}(\mathbf{x}) \quad (1)$$

where  $d_{out}$  and  $d_{in}$  represent the distances to the boundary from the fluid and obstacle domains, respectively. The resulting values are interpreted as follows:  $\Phi(\mathbf{x}) > 0$  indicates the fluid domain i.e., airflow,  $\Phi(\mathbf{x}) < 0$  signifies the interior of a building, and  $\Phi(\mathbf{x}) = 0$  marks the building boundary. For numerical stability,  $\Phi(\mathbf{x})$  is normalized to the range  $[-1, 1]$  before being used as a model input.

## 2.3 Sparse Observation and Interpolation Preprocessing

To simulate the constraints inherent in real-world data acquisition, a sparse sensor configuration is implemented by randomly sampling the ground-truth flow field at a 5% mask ratio. This sampling process is restricted to the fluid domain, where a binary sensor mask is generated to distinguish between observed and unobserved coordinates. These sparse measurements are subsequently processed through a pipeline designed to provide a dense initial estimate of the flow structure.

First, a continuous velocity field is synthesized by applying Delaunay triangulation-based linear interpolation to the sampled sparse points. For grid points situated outside the convex hull of the sensor locations, nearest-neighbor interpolation is utilized to augment the field and ensure spatial continuity across the entire  $128 \times 128$  domain. Finally, the velocity components are scaled via global min-max normalization based on training set statistics. This normalization process ensures numerical stability and facilitates efficient convergence during the subsequent training of the deep learning models.

## 2.4 Input Composition

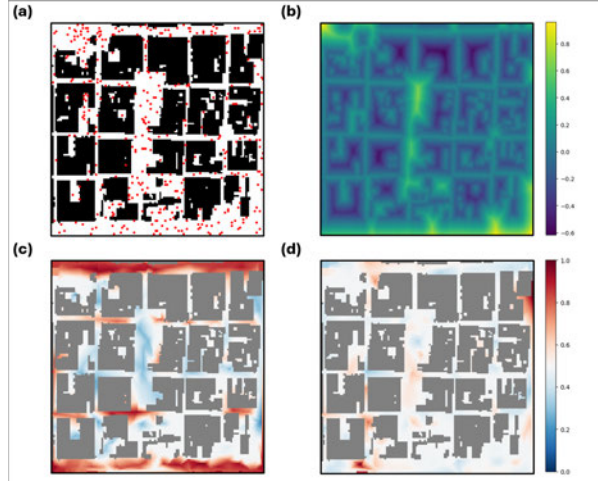


Figure 1: Contours of input with  $128 \times 128$  sizes. (a) 5% sparse points; (b) SDF field; (c) Interpolated  $u$  field; (d) Interpolated  $v$  field.

As shown in Figure 1 the input of models is a multi-channel tensor  $\mathbf{X}_{in} \in \mathbb{R}^{4 \times H \times W}$  consisting of: (a) a binary sensor mask, (b) the normalized SDF, (c) the interpolated  $u$ -velocity, and (d) the interpolated

$v$ -velocity. This composition allows the models to leverage both local sparse measurements and global geometric constraints.

## 2.5 Data Partitioning

The processed dataset is partitioned into training, validation, and test sets using a ratio of 8:1:1. This split ensures that the models are trained on a substantial diversity of flow features while maintaining sufficient independent data for hyperparameter tuning and final performance evaluation. All partitions are drawn from the realistic urban domain to evaluate the ability of the models to generalize across complex, authentic architectural layouts.

The training set is utilized for optimizing model parameters, while the validation set serves as a criterion for early stopping to prevent overfitting. The final evaluation is performed on the test set, comparing the reconstructed high-resolution flow fields against the ground-truth LES data. This partitioning strategy provides a rigorous framework for assessing the recovery of high-frequency turbulent patterns and building wake interactions under sparse observation constraints.

## 3 Methodology

This section presents our approach for reconstructing dense urban flow fields from sparse sensor observations. We formulate the problem as a conditional generation task and introduce a diffusion-based framework specifically designed for turbulent flow reconstruction.

### 3.1 Problem Formulation

Given sparse velocity measurements and urban geometry information, our goal is to reconstruct the full velocity field over the computational domain. Formally, let  $\mathbf{x} = (u, v) \in \mathbb{R}^{2 \times H \times W}$  denote the ground-truth velocity field, where  $u$  and  $v$  represent the horizontal and vertical velocity components, respectively. The conditioning input  $\mathbf{c} \in \mathbb{R}^{4 \times H \times W}$  consists of four channels:

$$\mathbf{c} = [\tilde{u}, \tilde{v}, \mathbf{m}, \Phi] \quad (2)$$

where  $\tilde{u}$  and  $\tilde{v}$  are the linearly interpolated velocity fields from sparse observations,  $\mathbf{m} \in \{0, 1\}^{H \times W}$  is the binary sensor mask indicating observation locations, and  $\Phi$  is the normalized signed distance function encoding building geometry.

### 3.2 Baseline: Deterministic U-Net

As a baseline, we employ a standard U-Net architecture Ronneberger et al. [2015] for deterministic regression. The encoder-decoder structure with skip connections directly maps the conditioning input to the predicted velocity field. The network is trained to minimize the masked mean squared error:

$$\mathcal{L}_{\text{UNet}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|^2 \quad (3)$$

where  $\mathcal{M}$  denotes the set of valid fluid domain pixels. While effective for smooth field reconstruction, deterministic regression inherently suffers from spectral bias—the tendency to average over possible solutions, resulting in over-smoothed predictions that underestimate high-frequency turbulent fluctuations Fukami et al. [2019].

### 3.3 Diffusion-based Flow Reconstruction

To address the spectral limitations of deterministic methods, we propose a conditional diffusion model Ho et al. [2020] that learns the full conditional distribution  $p(\mathbf{x}|\mathbf{c})$  rather than a point estimate. This generative approach is particularly well-suited for turbulent flow reconstruction for three reasons: (1) turbulence is inherently stochastic, making probabilistic modeling natural; (2) diffusion models excel at generating fine-grained details through iterative refinement; and (3) the conditioning mechanism allows seamless integration of sparse observations and geometric constraints.

Our model builds upon the denoising diffusion probabilistic model (DDPM) framework with several adaptations for flow field data. We adopt a cosine noise schedule Nichol and Dhariwal [2021] for

smoother noise levels across timesteps, and employ the velocity (v-prediction) parameterization Salimans and Ho [2022]:

$$\mathbf{v}_t = \sqrt{\bar{\alpha}_t} \boldsymbol{\epsilon} - \sqrt{1 - \bar{\alpha}_t} \mathbf{x}_0 \quad (4)$$

where  $\bar{\alpha}_t$  is the cumulative noise schedule and  $\boldsymbol{\epsilon}$  is the added noise. This parameterization provides more stable gradients across all noise levels compared to direct noise prediction, which is beneficial for continuous physical fields.

The denoising network  $\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$  is implemented as a conditional U-Net that processes the concatenated noisy sample and conditioning input. Timestep information is injected via adaptive group normalization (AdaGN) Wu and He [2018], Dhariwal and Nichol [2021], where learned projections of sinusoidal timestep embeddings Vaswani et al. [2017] modulate the normalized features. A self-attention layer at the bottleneck captures long-range spatial dependencies crucial for modeling building wake interactions. The model is trained with a combined objective:

$$\mathcal{L} = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\|\hat{\mathbf{v}}_t - \mathbf{v}_t\|^2 \cdot \mathbf{m}] + \lambda \|\hat{\mathbf{x}}_0 - \mathbf{x}_0\|^2 \cdot \mathbf{m} \quad (5)$$

where the auxiliary reconstruction term ( $\lambda = 0.1$ ) improves sample quality. For efficient inference, we employ DDIM sampling Song et al. [2020] with 25 steps, providing a  $4\times$  speedup over the 100-step training process.

### 3.4 Key Design Choices for Urban Flow

Beyond the standard diffusion framework, we introduce two design choices specifically motivated by the characteristics of urban flow reconstruction.

**Interpolation-based Input Encoding.** A naive approach to sparse sensor reconstruction would directly encode point measurements and their coordinates as network inputs. However, this creates a fundamental mismatch with convolutional architectures, which are designed to exploit local spatial correlations in structured 2D grids. Transforming sparse 1D point data into dense 2D fields requires the network to first learn an implicit interpolation mapping before performing the actual flow reconstruction—an unnecessarily complex task.

Instead, we provide the network with linearly interpolated velocity fields  $(\tilde{u}, \tilde{v})$  as input channels. This design choice offers two advantages: (1) it aligns the input structure with the inductive bias of convolutional networks, enabling efficient exploitation of spatial locality; and (2) it shifts the learning objective from full-field generation to residual refinement, as the coarse flow structure is already present in the interpolated input. Combined with the binary sensor mask  $\mathbf{m}$  that indicates observation locations, the network can distinguish between reliable measurements and interpolated estimates.

**Residual Diffusion Formulation.** Urban flow fields exhibit strong spatial correlations that simple linear interpolation can partially capture. Rather than learning the entire flow field from scratch, we reformulate the task as residual prediction Jeon et al. [2024]:

$$\Delta \mathbf{x} = \mathbf{x} - \tilde{\mathbf{x}}_{\text{interp}}, \quad \hat{\mathbf{x}} = \tilde{\mathbf{x}}_{\text{interp}} + \Delta \hat{\mathbf{x}} \quad (6)$$

This formulation allows the diffusion model to focus on high-frequency corrections—precisely where deterministic interpolation fails—while leveraging the coarse structure already present in the input. The residuals have smaller magnitude and more consistent statistics, simplifying the learning task and improving training stability.

## 4 Results

### 4.1 Experimental Setup and Metrics

The proposed framework was evaluated on the *UrbanTales* dataset using a Sim-to-Real protocol to ensure robustness against sparse and noisy sensor data. We benchmarked our approach against three methodologies: Linear Interpolation (baseline), a deterministic U-Net, and our proposed Full-Image Diffusion model (**Diffusion**).

Performance was quantified using three metrics across flow speed ( $|U|$ ) and velocity components ( $u, v$ ): Coefficient of Determination ( $R^2$ ) for predictive power, Normalized Root Mean Square Error (NRMSE), and Mean Absolute Error (MAE) for deviation accuracy. These metrics provide a comprehensive view of reconstruction fidelity, capturing both global trends and local variations.

### 4.2 Qualitative Analysis

We visually analyze the reconstruction quality using two representative datasets: **Case 1** (Standard Urban Layout, Singapore) and **Case 2** (Complex Dense Layout, Ukraine).

Figure 2 compares the global flow field reconstruction. The Linear baseline fails to capture non-linear dynamics, yielding disjointed fields. Similarly, the deterministic U-Net produces blurred vector fields, struggling with sharp discontinuities near buildings. In contrast, the **Diffusion** model accurately recovers the overall flow direction and magnitude, maintaining structural consistency in both scenarios.

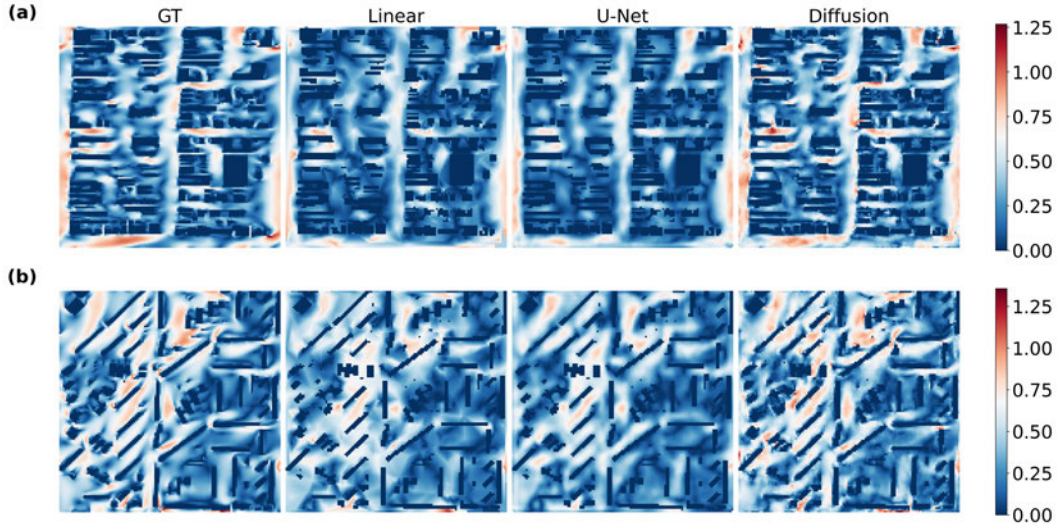


Figure 2: **Global Reconstruction Comparison.** Visual comparison of Ground Truth vs. Model Predictions for (a) Case 1 and (b) Case 2. The Diffusion model exhibits sharper flow boundaries and more accurate global structures compared to the blurred outputs of baselines.

The advantage of the proposed method is most evident in fine-scale structures (Figure 3). U-Net tends to smooth out high-frequency fluctuations due to its MSE-based objective, acting as a low-pass filter Fukami et al. [2019]. Conversely, the **Diffusion** model successfully reconstructs sharp gradients and vortex shedding patterns. This demonstrates its generative capability to recover intricate **local details** lost in sparse observations, effectively overcoming the spectral bias of deterministic regression.



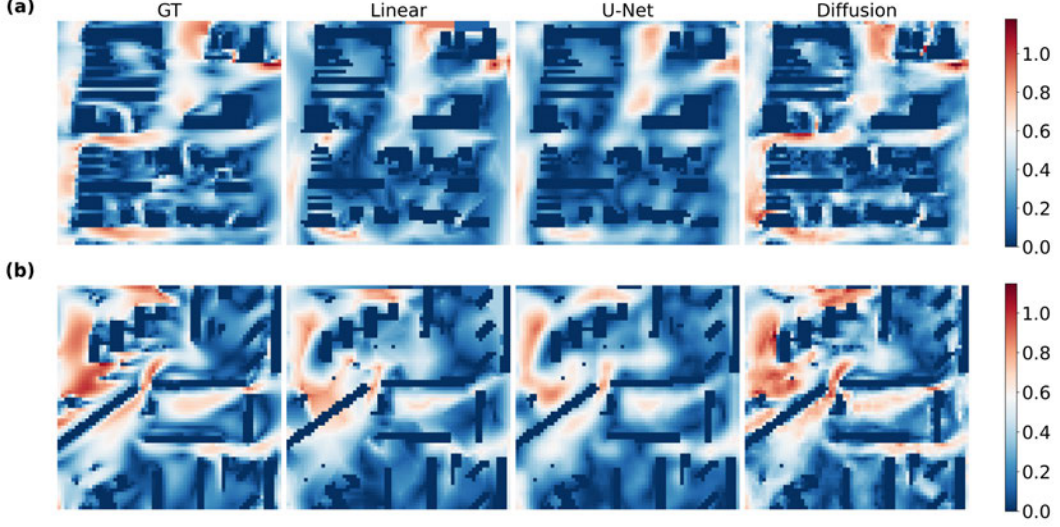


Figure 3: **Detail Preservation Analysis (Zoom-in)**. Zoomed-in views of the top-right quadrant. Diffusion preserves sharp gradients and small-scale turbulent structures, whereas U-Net tends to blur these features.

### 4.3 Spectral Consistency Analysis

To validate physical consistency, we analyzed the isotropic kinetic energy spectrum,  $E(k)$ . As shown in Figure 4, regression-based baselines (Linear, U-Net) suffer from significant spectral decay at high wavenumbers, failing to capture the turbulence energy cascade Callaham et al. [2019].

The **Diffusion** model, however, aligns closely with the Ground Truth spectrum across all scales. By iteratively refining noise, the diffusion process injects high-frequency details matching the target distribution, confirming that our generative framework effectively mitigates the smoothing artifacts common in deep learning-based fluid reconstruction Yousif et al. [2023].

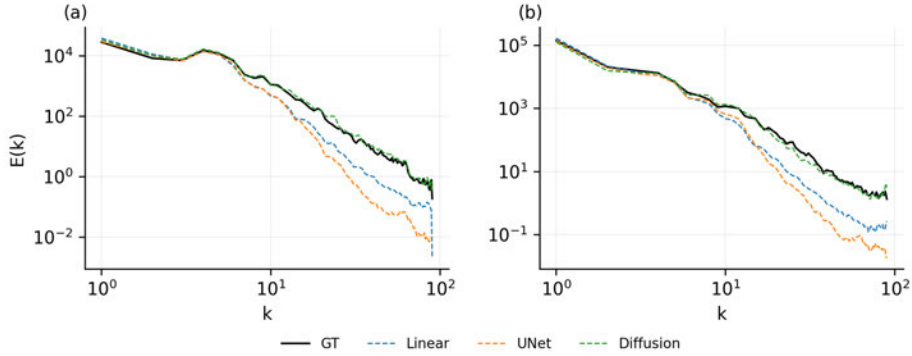


Figure 4: **Isotropic Kinetic Energy Spectrum**. Comparison of spectral energy density for (a) Case 1 and (b) Case 2. The Diffusion model (red) closely follows the Ground Truth (black) at high wavenumbers, indicating superior preservation of small-scale turbulence energy compared to baselines.

### 4.4 Quantitative Comparison

Table 1 summarizes the quantitative performance for representative cases and the entire dataset. The **Diffusion** model consistently outperforms baselines in terms of  $R^2$ , NRMSE, and MAE. Notably, the improvement is substantial in the dense urban layout (Case 2) and the overall dataset. For instance,



the Diffusion model achieves an aggregate  $R^2$  of 0.571 for flow speed, significantly surpassing the 0.449 of U-Net, demonstrating robust generalization under sparse sensing conditions.

Table 1: **Quantitative Metrics Comparison.** Evaluation on representative cases and the entire dataset (mean values). Best results are highlighted in **bold**.

| Dataset  | Model            | Speed ( $ U $ ) |                    |                  | $u$ -velocity  |                    |                  | $v$ -velocity  |                    |                  |
|----------|------------------|-----------------|--------------------|------------------|----------------|--------------------|------------------|----------------|--------------------|------------------|
|          |                  | $R^2 \uparrow$  | NRMSE $\downarrow$ | MAE $\downarrow$ | $R^2 \uparrow$ | NRMSE $\downarrow$ | MAE $\downarrow$ | $R^2 \uparrow$ | NRMSE $\downarrow$ | MAE $\downarrow$ |
| Case 1   | Linear           | 0.479           | 0.129              | 0.113            | 0.760          | 0.094              | 0.103            | 0.557          | 0.089              | 0.108            |
|          | U-Net            | 0.540           | 0.122              | 0.111            | 0.789          | 0.088              | 0.103            | 0.630          | 0.081              | 0.104            |
|          | <b>Diffusion</b> | <b>0.671</b>    | <b>0.103</b>       | <b>0.091</b>     | <b>0.835</b>   | <b>0.078</b>       | <b>0.087</b>     | <b>0.705</b>   | <b>0.072</b>       | <b>0.091</b>     |
| Case 2   | Linear           | 0.445           | 0.137              | 0.116            | 0.670          | 0.096              | 0.109            | 0.591          | 0.072              | 0.111            |
|          | U-Net            | 0.489           | 0.132              | 0.118            | 0.710          | 0.090              | 0.109            | 0.641          | 0.068              | 0.114            |
|          | <b>Diffusion</b> | <b>0.642</b>    | <b>0.110</b>       | <b>0.094</b>     | <b>0.764</b>   | <b>0.081</b>       | <b>0.097</b>     | <b>0.734</b>   | <b>0.058</b>       | <b>0.094</b>     |
| All Data | Linear           | 0.434           | 0.145              | 0.128            | 0.621          | 0.091              | 0.122            | 0.565          | 0.083              | 0.114            |
|          | U-Net            | 0.449           | 0.143              | 0.132            | 0.642          | 0.087              | 0.123            | 0.599          | 0.080              | 0.116            |
|          | <b>Diffusion</b> | <b>0.571</b>    | <b>0.124</b>       | <b>0.111</b>     | <b>0.658</b>   | <b>0.081</b>       | <b>0.110</b>     | <b>0.601</b>   | <b>0.076</b>       | <b>0.108</b>     |

## 5 Conclusion

In this study, we presented a novel coarse-to-fine framework for reconstructing high-resolution urban flow fields from extremely sparse sensor measurements (5% sparsity). By integrating a conditional Denoising Diffusion Probabilistic Model (DDPM) with DDIM sampling, we effectively addressed the ill-posed nature of the sparse reconstruction problem. Our extensive experimental results demonstrate that the synergy between deterministic interpolation and stochastic generation is crucial for achieving high-fidelity reconstruction. Linear interpolation serves as a robust conditioner, providing a foundational mean flow field, but it inherently fails to capture the non-linear fluid dynamics. The proposed Diffusion model successfully bridges this gap by treating the interpolated field as a structural guide, thereby focusing its generative capacity on recovering the high-frequency residual details.

The core novelty of this work lies in demonstrating that, leveraging the unique expressive power of diffusion models, it is possible to successfully reconstruct **local details** across the entire domain even with extremely **sparse input data**. While traditional regression-based baselines produce over-smoothed predictions that fail to capture the physics of turbulence, our model excels at capturing intricate structures such as complex wake vortices and shear layers. This capability was rigorously validated not only through standard error metrics but also via spectral analysis, confirming the preservation of energy across scales.

Despite these significant advancements, the current study has limitations that open avenues for future research. First, regarding inference efficiency, we adopted DDIM sampling to accelerate the generation process. However, the absolute reconstruction accuracy ( $R^2 \approx 0.6$ ) is fundamentally limited by the **extreme information bottleneck** imposed by the 5% sparsity setting, rather than the sampling scheme itself. Second, the current framework operates on 2D static snapshots, neglecting the temporal evolution and 3D vertical interactions that are inherent in real-world urban flows.

To address these challenges, future work will focus on incorporating physics-informed constraints, such as Navier-Stokes residuals and divergence-free losses, directly into the training process to enforce fluid-dynamic consistency Raissi et al. [2019], Karniadakis et al. [2021]. Furthermore, extending the architecture to spatiotemporal models (e.g., Video Diffusion) will be essential to capture volumetric flow dynamics and provide a more comprehensive tool for real-time environmental monitoring Wang et al. [2020].

## A Implementation Details

**Computational Resources.** All experiments were conducted on a single NVIDIA GeForce RTX 3090 GPU (24GB VRAM). Training the conditional diffusion model took approximately 20 minutes.

## References

- Jared L Callaham, Kazuki Maeda, and Steven L Brunton. Robust flow reconstruction from limited measurements via sparse representation. *Physical Review Fluids*, 4(10):103907, 2019.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.
- Kai Fukami, Koji Fukagata, and Kunihiro Taira. Super-resolution reconstruction of turbulent flows with machine learning. *Journal of Fluid Mechanics*, 870:106–120, 2019.
- Kai Fukami, Koji Fukagata, and Kunihiro Taira. Assessment of supervised machine learning methods for fluid flows. *Theoretical and Computational Fluid Dynamics*, 34(4):497–519, February 2020. ISSN 1432-2250. doi: 10.1007/s00162-020-00518-y. URL <http://dx.doi.org/10.1007/s00162-020-00518-y>.
- Huanxiang Gao, Gang Hu, Dongqin Zhang, Wenjun Jiang, KT Tse, KCS Kwok, and Ahsan Kareem. Urban wind field prediction based on sparse sensors and physics-informed graph-assisted auto-encoder. *Computer-Aided Civil and Infrastructure Engineering*, 39(10):1409–1430, 2024.
- C García-Sánchez, J Van Beeck, and C Gorlé. Predictive large eddy simulations for urban flows: Challenges and opportunities. *Building and Environment*, 139:146–156, 2018.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- Joongoo Jeon, Juhyeon Lee, Ricardo Vinuesa, and Sung Joong Kim. Residual-based physics-informed transfer learning: A hybrid method for accelerating long-term cfd simulations via deep learning. *International Journal of Heat and Mass Transfer*, 220:124900, 2024.
- Javier Jiménez. Computing high-reynolds-number turbulence: will simulations ever replace experiments? *Journal of Turbulence*, 4(1):022, 2003.
- George Em Karniadakis, Ioannis G Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021.
- Patrick Kastner and Timur Dogan. A gan-based surrogate model for instantaneous urban wind flow prediction. *Building and Environment*, 242:110384, 2023.
- Sangseung Lee and Donghyun You. Data-driven prediction of unsteady flow over a circular cylinder using deep learning. *Journal of Fluid Mechanics*, 879:217–254, 2019.
- Mohammad Mortezaazadeh, Liangzhu Leon Wang, Maher Albettar, and Senwen Yang. Cityffd—city fast fluid dynamics for urban microclimate simulations on graphics processing units. *Urban Climate*, 41:101063, 2022.
- Negin Nazarian, Jiachen Lu, Mathew J Lipson, Melissa A Hart, Sijie Liu, E Scott Krayenhoff, Lewis Blunn, and Alberto Martilli. Urbantales: A large-eddy simulation dataset for urban canopy layer turbulence and parameterization. *Bulletin of the American Meteorological Society*, 106(12): E2461–E2478, 2025.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International conference on machine learning*, pages 8162–8171. PMLR, 2021.

- Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
- Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models. *arXiv preprint arXiv:2202.00512*, 2022.
- Xuqiang Shao, Zhijian Liu, Siqi Zhang, Zijia Zhao, and Chenxing Hu. Pignn-cfd: A physics-informed graph neural network for rapid predicting urban wind field defined on unstructured mesh. *Building and Environment*, 232:110056, 2023.
- Heesoo Shin, Seyed Morteza Habibi Khorasani, Zhaoyu Shi, Jiasheng Yang, Shervin Bagheri, and Sangseung Lee. Data-driven discovery of drag-inducing elements on a rough surface through convolutional neural networks. *Physics of Fluids*, 36(9), 2024.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.
- Kunihiko Taira, Georgios Rigas, and Kai Fukami. Machine learning in fluid dynamics: A critical assessment. *Physical Review Fluids*, 10(9):090701, 2025.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Ricardo Vinuesa and Steven L Brunton. Enhancing computational fluid dynamics with machine learning. *Nature Computational Science*, 2(6):358–366, 2022.
- Han Wang, Akram Murid, M Reza Amini, Andrew Thelen, Hamed Ben Gida, and Michael Roster. Physics-informed deep super-resolution of spatiotemporal turbulent flows. *Journal of Computational Physics*, 408:109296, 2020.
- Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European conference on computer vision (ECCV)*, pages 3–19, 2018.
- M Z Yousif, L Yu, and H Lim. High-fidelity reconstruction of turbulent flow from sparse measurements using generative adversarial networks. *Journal of Fluid Mechanics*, 957:A26, 2023.

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